

CARE TRANSITION EFFICIENCY & PLACEMENT OUTCOME ANALYTICS

A Business Analytics Project Report

**Submitted in Partial Fulfillment of the Requirements for the Award of the
CERTIFICATE of INTERNSHIP**

Specialization: Business Analytics

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Submitted To

Unified Mentor Pvt. Ltd.

Project Organization

Unified Mentor

Department of Health and Human Services (HHS)

United States Government

CERTIFICATE

This is to certify that the project report entitled "**Care Transition Efficiency & Placement Outcome Analytics**" has been successfully completed by **Ashutosh sharma**, an intern of **MBA (Business Analytics) at Unified Mentor Pvt. Ltd.**, as a part of the academic curriculum prescribed for the Master of Business Administration degree.

The work presented in this report is original and has been carried out under proper academic guidance. To the best of our knowledge, this project has not been submitted previously, either in full or in part, for the award of any other degree or diploma.

The project demonstrates the application of Business Analytics concepts, Python programming, Streamlit dashboard development, and data visualization techniques in analyzing operational efficiency within the Unaccompanied Alien Children (UAC) care transition process.

Faculty Guide

Signature: _____

Date: _____

DECLARATION

I hereby declare that the project entitled "**Care Transition Efficiency & Placement Outcome Analytics**" submitted in partial fulfillment of the requirements for the certificate of **internship (Business Analytics)** is my original work.

The analysis, dashboard development, visualizations, and interpretations presented in this report have been carried out by me using publicly available data and open-source analytical tools. Wherever reference has been made to the work of others, proper acknowledgment has been given.

I further declare that this project has not been submitted previously to any university or institution for the award of any degree or diploma.

Submitted By

Ashutosh Sharma

Date: _____

ACKNOWLEDGEMENT

I express my sincere gratitude to **Amity University Online** for providing me with the opportunity to undertake this Business Analytics project as part of my MBA curriculum.

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Special thanks are extended to the open-source developer community for providing powerful analytical libraries including **Python, Pandas, NumPy, Matplotlib, Plotly, and Streamlit**, which enabled the successful implementation of this project.

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Ashutosh Sharma

ABSTRACT

The Unaccompanied Alien Children (UAC) Program represents a multi-stage operational care pipeline responsible for receiving, sheltering, managing, and reunifying children with approved sponsors. While operational reports generally focus on monitoring the number of children under custody, limited attention has been given to evaluating the efficiency of care transitions and operational performance across the complete pipeline.

This project introduces a Business Analytics framework designed to evaluate the operational efficiency of care transitions by analyzing daily movement between Customs and Border Protection (CBP) custody, Health and Human Services (HHS) care facilities, and successful sponsor placement. The project focuses on transforming traditional operational reporting into actionable business intelligence through process-oriented performance metrics and interactive dashboard visualization.

The study utilizes Python-based data analysis techniques including data preprocessing, feature engineering, exploratory data analysis, statistical performance measurement, and dashboard development. Multiple Key Performance Indicators (KPIs) were developed, including the Transfer Efficiency Ratio, Discharge Effectiveness Index, Pipeline Throughput Rate, CBP Backlog, HHS Backlog, and Outcome Stability Score. These indicators enable continuous monitoring of operational efficiency and assist decision-makers in identifying process bottlenecks and delays.

An interactive Streamlit dashboard was developed to provide dynamic visualization of operational trends, KPI monitoring, backlog detection, and placement outcomes. The dashboard includes interactive filters, trend analysis, comparative charts, and performance indicators that facilitate real-time operational monitoring.

The findings indicate that process-oriented analytics significantly improve visibility into operational performance compared to traditional aggregate reporting. By identifying delays, measuring throughput efficiency, and monitoring placement outcomes, the proposed analytical framework provides valuable insights for improving resource allocation, reducing operational bottlenecks, and accelerating child reunification processes.

Overall, this project demonstrates how Business Analytics techniques combined with interactive dashboard technologies can transform operational datasets into actionable decision-support systems. The developed framework contributes toward improving transparency, operational efficiency, and evidence-based policy formulation within the UAC care transition process.

Keywords: Business Analytics, Care Transition, UAC Program, HHS, CBP, Streamlit Dashboard, Python, Data Visualization, KPI Analysis, Operational Efficiency, Process Analytics, Placement Outcomes.

EXECUTIVE SUMMARY

The **Care Transition Efficiency & Placement Outcome Analytics** project was undertaken to evaluate and improve the operational efficiency of the **Unaccompanied Alien Children (UAC) Program**, administered by the **U.S. Department of Health and Human Services (HHS)**. The program is responsible for managing the complete care transition pipeline of unaccompanied children, beginning with their apprehension by **Customs and Border Protection (CBP)** and ending with their successful reunification through approved sponsors.

Traditional reporting systems primarily monitor the number of children present at different stages of the care process. While these reports provide information regarding operational workload, they do not adequately measure process efficiency, transition speed, backlog accumulation, or placement outcomes. As a result, administrators often face challenges in identifying bottlenecks, evaluating operational performance, and implementing timely corrective actions.

The objective of this project was to transform traditional operational reporting into an analytical decision-support framework using Business Analytics techniques. The project focused on measuring operational efficiency through the development of Key Performance Indicators (KPIs), performing exploratory data analysis, and designing an interactive Streamlit dashboard that enables continuous monitoring of care transitions.

The project utilized a structured analytical methodology beginning with data collection, preprocessing, feature engineering, exploratory data analysis, KPI calculation, dashboard development, and deployment. Python served as the primary programming language, while libraries including Pandas, NumPy, Matplotlib, Plotly, and Streamlit were used for data processing, visualization, and dashboard creation. Version control was maintained through GitHub, and the final dashboard was deployed using Streamlit Community Cloud to provide online accessibility.

Several operational KPIs were developed to evaluate the performance of the care transition pipeline. These included the **Transfer Efficiency Ratio**, which measures the effectiveness of transitions from CBP custody to HHS care; the **Discharge Effectiveness Index**, which evaluates sponsor placement performance; the **Pipeline Throughput Rate**, which measures the movement of children through the system; the **CBP Backlog** and **HHS Backlog**, which identify workload accumulation at different stages; and the **Outcome Stability Score**, which evaluates consistency in placement outcomes over time.

The exploratory data analysis identified significant operational patterns and trends. Daily apprehension levels varied throughout the reporting period, directly influencing the workload experienced by CBP and HHS facilities. The Transfer Efficiency Ratio remained relatively stable for most of the reporting period, indicating effective movement between operational stages. However, temporary increases in CBP backlog highlighted periods during which transfers could not keep pace with incoming cases. Similarly, variations in discharge effectiveness reflected fluctuations in sponsor placement activities, emphasizing the need for continuous monitoring and resource optimization.

An interactive dashboard was developed using Streamlit to provide real-time visualization of operational performance. The dashboard includes KPI summary cards, trend analyses, backlog monitoring, comparative charts, and interactive date filters. This user-friendly interface enables decision-makers to monitor operational efficiency dynamically and identify emerging issues without relying solely on static reports.

Based on the findings, several recommendations have been proposed. These include strengthening coordination between CBP and HHS, implementing automated KPI monitoring systems, improving sponsor verification workflows, expanding dashboard-based reporting, and adopting predictive analytics for future operational planning. These recommendations aim to reduce delays, improve resource utilization, and enhance overall operational transparency.

The project demonstrates how Business Analytics can convert raw operational data into actionable managerial insights. By combining data preprocessing, feature engineering,

exploratory data analysis, KPI development, and interactive visualization, the project provides a comprehensive framework for evaluating care transition efficiency. The analytical model developed through this study can support evidence-based decision-making and may serve as a foundation for future predictive and AI-driven operational management systems.

In conclusion, the **Care Transition Efficiency & Placement Outcome Analytics** project successfully fulfilled its objectives by developing a comprehensive analytical framework capable of monitoring operational performance, identifying bottlenecks, and supporting faster and more efficient care transitions. The project demonstrates the practical application of Business Analytics in solving real-world public sector challenges and highlights the importance of data-driven decision-making in improving humanitarian operations.

CHAPTER 1

INTRODUCTION

1.1 Introduction

The increasing adoption of Business Analytics across public administration has transformed the way organizations monitor operational performance, allocate resources, and make strategic decisions. Government agencies responsible for humanitarian services manage large volumes of operational data every day. However, collecting data alone is insufficient unless it is converted into meaningful insights that support effective decision-making.

The **Unaccompanied Alien Children (UAC) Program**, administered by the **U.S. Department of Health and Human Services (HHS)**, represents one such operational environment where data-driven decision-making can significantly improve efficiency and humanitarian outcomes. The program is responsible for providing temporary care, medical screening, shelter, education, case management, and eventual reunification of unaccompanied children with approved sponsors after their transfer from **Customs and Border Protection (CBP)** custody.

Traditionally, operational reports within the UAC Program have focused primarily on monitoring the number of children under CBP custody and HHS care. Although these statistics provide information regarding workload and capacity, they provide limited understanding of how efficiently children move through the complete care transition pipeline.

Business Analytics provides an opportunity to transform operational data into performance intelligence by measuring process efficiency, identifying operational bottlenecks, monitoring throughput, and evaluating placement outcomes. This project applies analytical methodologies to evaluate the complete care transition process rather than focusing solely on aggregate population statistics.

1.2 Background of the Study

The UAC Program operates as a structured operational pipeline involving multiple stages of care and transition. Children who enter the United States without a parent or legal guardian are initially apprehended by **Customs and Border Protection (CBP)**. Following initial processing, they are transferred to facilities managed by the **Office of Refugee Resettlement (ORR)** under the **Department of Health and Human Services (HHS)**.

Once admitted into HHS care, children receive medical examinations, counseling services, educational support, legal assistance, and case management until they can be safely reunited with qualified sponsors.

The operational stages of the pipeline include:

- Apprehension
- CBP Custody
- Transfer to HHS Care
- Medical Screening
- Shelter Services
- Case Management
- Sponsor Verification
- Discharge and Reunification

Every stage generates operational data that can be analyzed to evaluate process performance. Delays occurring at any stage affect subsequent operations and may increase overall care duration.

This project treats the UAC Program as an operational workflow rather than a simple population monitoring system. Consequently, greater emphasis is placed on transition efficiency, throughput, backlog management, and placement outcomes.

1.3 Business Analytics Perspective

Business Analytics refers to the systematic use of data, statistical analysis, predictive techniques, and visualization tools to improve organizational decision-making.

Modern organizations increasingly rely on Business Analytics to answer questions such as:

- How efficiently are processes operating?
- Where are operational bottlenecks occurring?
- Which activities require improvement?
- How can organizational performance be optimized?

Within the UAC Program, Business Analytics enables administrators to evaluate the effectiveness of operational transitions rather than merely reporting daily counts.

Python-based analytical tools, including Pandas, NumPy, Matplotlib, Plotly, and Streamlit, enable efficient processing of operational datasets while interactive dashboards improve decision-making through real-time visualization.

1.4 Need for the Study

Although detailed operational records are collected daily, limited analytical emphasis has been placed on evaluating care transition efficiency.

The study was undertaken because several operational questions remain unanswered through conventional reporting.

These include:

- How efficiently are children transferred from CBP to HHS?
- Are discharge rates keeping pace with new arrivals?
- Which periods experience operational congestion?
- How consistent are placement outcomes?
- Are operational improvements occurring over time?

Traditional reports provide numerical summaries but fail to explain why operational delays occur or how process efficiency can be improved.

This study addresses these limitations by developing performance indicators capable of measuring operational efficiency throughout the care pipeline.

1.5 Statement of the Problem

Existing reporting systems primarily monitor aggregate population statistics rather than operational performance.

As a result:

- Process bottlenecks remain difficult to identify.
- Resource planning becomes reactive instead of proactive.
- Operational delays remain hidden.
- Placement efficiency cannot be measured objectively.
- Performance comparisons across time become difficult.

The absence of standardized analytical indicators limits evidence-based policy formulation.

This project develops measurable Key Performance Indicators (KPIs) that evaluate operational performance and support informed decision-making.

1.6 Purpose of the Study

The purpose of this study is to evaluate the operational efficiency of the UAC care transition pipeline using Business Analytics methodologies.

The study seeks to transform operational data into meaningful managerial insights through data preprocessing, KPI development, visualization, and interactive dashboard reporting.

By focusing on operational transitions rather than population counts alone, the project contributes toward improved transparency and operational management.

1.7 Scope of the Study

The scope of this project includes analysis of daily operational data representing different stages of the UAC care transition process.

The study covers:

- Daily apprehensions
- CBP custody
- Transfers to HHS
- HHS care population
- Sponsor placements
- Operational backlogs
- Throughput analysis
- Trend analysis

- KPI development
- Interactive dashboard creation

The project focuses on descriptive and diagnostic analytics rather than predictive modeling.

1.8 Significance of the Study

The study provides value to multiple stakeholders.

For Government Agencies

Provides greater operational transparency and performance monitoring.

For Decision Makers

Supports evidence-based policy formulation through KPI-driven insights.

For Case Managers

Helps identify workflow delays and improve reunification timelines.

For Researchers

Demonstrates practical application of Business Analytics in public administration.

For Students

Provides a real-world implementation of Python, Streamlit, and Business Intelligence techniques.

1.9 Research Objectives

The major objectives of the study are:

Primary Objectives

- Measure CBP to HHS Transfer Efficiency.
- Evaluate Discharge Effectiveness.

- Measure Pipeline Throughput.
- Identify Operational Bottlenecks.
- Monitor Backlog Accumulation.

Secondary Objectives

- Improve operational visibility.
 - Support faster reunification.
 - Enhance case management.
 - Develop an interactive dashboard.
 - Generate actionable business insights.
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1.10 Research Questions

The study attempts to answer the following research questions:

1. How efficiently are children transferred from CBP custody to HHS care?
 2. Are sponsor placements keeping pace with incoming cases?
 3. During which periods do operational backlogs increase?
 4. Which operational indicators best measure care transition efficiency?
 5. How can Business Analytics improve operational decision-making?
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1.11 Tools and Technologies Used

The following technologies were used throughout the project:

Tool	Purpose
Python	Data Analysis
Pandas	Data Cleaning

NumPy	Numerical Analysis
Matplotlib	Data Visualization
Plotly	Interactive Charts
Streamlit	Dashboard Development
VS Code	Development Environment
GitHub	Version Control
Streamlit Community Cloud	Dashboard Deployment

1.12 Project Workflow

The project followed the workflow shown below:

Raw Dataset

↓

Data Cleaning

↓

Feature Engineering

↓

KPI Calculation



Exploratory Data Analysis



Visualization



Dashboard Development



Business Insights



Recommendations

1.13 Chapter Summary

This chapter introduced the Care Transition Efficiency & Placement Outcome Analytics project and discussed the operational background of the UAC Program. The chapter highlighted the importance of Business Analytics in evaluating operational performance beyond traditional population monitoring.

The study's objectives, scope, significance, research questions, tools, and workflow were also presented. These concepts establish the foundation for the subsequent chapters, where existing literature, dataset characteristics, analytical methodology, dashboard development, findings, and recommendations are discussed in detail.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

A literature review is a systematic examination of existing research, academic publications, government reports, and industry practices related to the study topic. It establishes the theoretical foundation of the research and identifies gaps that justify the current study.

This project combines concepts from **Business Analytics**, **Process Analytics**, **Public Sector Performance Management**, **Key Performance Indicators (KPIs)**, and **Interactive Dashboard Development**. The literature reviewed demonstrates how analytical methods have evolved from descriptive reporting to intelligent decision-support systems that enable organizations to improve efficiency, transparency, and operational effectiveness.

The objective of this chapter is to review existing knowledge related to operational analytics and explain how these concepts are applied to evaluate the efficiency of the Unaccompanied Alien Children (UAC) care transition process.

2.2 Business Analytics

Business Analytics is the practice of collecting, processing, analyzing, and interpreting data to support organizational decision-making. It enables organizations to convert raw operational data into meaningful information that improves efficiency and strategic planning.

Modern Business Analytics consists of four major categories:

Descriptive Analytics

Descriptive analytics explains what has happened by summarizing historical data through reports, dashboards, charts, and statistical measures.

Example:

Daily reporting of children admitted into CBP custody.

Diagnostic Analytics

Diagnostic analytics investigates why certain events occurred by comparing trends, identifying anomalies, and analyzing relationships among variables.

Example:

Determining why discharge rates decreased during a particular month.

Predictive Analytics

Predictive analytics uses statistical models and machine learning techniques to forecast future outcomes based on historical data.

Example:

Forecasting future care demand based on historical apprehension patterns.

Prescriptive Analytics

Prescriptive analytics recommends actions that optimize organizational performance using analytical models.

Example:

Suggesting resource allocation strategies to reduce operational backlogs.

The current project primarily focuses on **Descriptive Analytics** and **Diagnostic Analytics**, while also providing a foundation for future predictive applications.

2.3 Process Analytics

Process Analytics evaluates how efficiently work moves through organizational processes rather than simply measuring outputs.

Unlike traditional reporting, Process Analytics emphasizes:

- Process efficiency
- Transition speed
- Waiting time
- Operational bottlenecks
- Resource utilization
- Throughput

The UAC care system functions as a multi-stage operational pipeline where each stage directly affects subsequent activities. Measuring transition efficiency therefore becomes essential for improving overall operational performance.

The analytical framework developed in this project evaluates each stage of the care transition pipeline using measurable performance indicators.

2.4 Key Performance Indicators (KPIs)

Key Performance Indicators (KPIs) are measurable values that indicate how effectively an organization achieves its objectives.

Effective KPIs possess the following characteristics:

- Quantifiable
- Objective
- Actionable
- Comparable
- Easy to interpret

For operational process evaluation, KPIs enable administrators to monitor efficiency continuously and detect deviations before they become significant operational issues.

The following KPIs were developed in this project:

- Transfer Efficiency Ratio
- Discharge Effectiveness Index
- Pipeline Throughput Rate
- CBP Backlog
- HHS Backlog
- Outcome Stability Score

These indicators collectively provide a comprehensive evaluation of operational performance.

2.5 Public Sector Performance Analytics

Government organizations increasingly rely on analytical systems to improve accountability, transparency, and service delivery.

Unlike private organizations, public institutions must balance efficiency with humanitarian and social objectives. Consequently, performance measurement must extend beyond financial indicators and incorporate operational effectiveness.

Performance analytics within public administration supports:

- Resource planning
- Policy evaluation
- Performance monitoring
- Risk identification
- Continuous improvement

The UAC Program represents an ideal application of public sector analytics because operational efficiency directly influences humanitarian outcomes.

2.6 Data Visualization and Dashboards

Data visualization converts numerical information into graphical representations that simplify interpretation and support decision-making.

Effective dashboards enable users to:

- Monitor performance
- Detect trends
- Compare operational indicators
- Identify bottlenecks
- Evaluate organizational efficiency

Interactive dashboards provide significantly greater value than static reports because users can dynamically explore different time periods and operational conditions.

This project employs **Streamlit** to develop an interactive dashboard incorporating KPI cards, trend analysis, operational charts, and date filtering.

2.7 Python for Business Analytics

Python has become one of the most widely adopted programming languages for Business Analytics due to its simplicity, flexibility, and extensive analytical ecosystem.

The project utilizes several Python libraries:

Pandas

Used for data loading, preprocessing, feature engineering, and manipulation.

NumPy

Provides numerical computation capabilities required for analytical calculations.

Matplotlib

Generates static analytical visualizations.

Plotly

Creates interactive charts that improve dashboard usability.

Streamlit

Develops interactive web applications without requiring extensive web development knowledge.

The integration of these libraries provides a complete analytical environment for operational performance evaluation.

2.8 Previous Research

Previous studies in operational analytics have demonstrated that organizations benefit significantly from monitoring process efficiency rather than relying solely on aggregate statistics.

Research in healthcare logistics, supply chain management, humanitarian operations, and government administration has consistently shown that KPI-based monitoring improves organizational performance by:

- Identifying delays
- Reducing operational bottlenecks
- Improving throughput
- Supporting strategic decision-making

However, relatively limited research has focused specifically on analytical evaluation of the UAC care transition process.

This project contributes by applying Business Analytics principles to evaluate operational efficiency within this humanitarian care pipeline.

2.9 Research Gap

Existing operational reporting primarily emphasizes population counts rather than transition efficiency.

The following research gaps were identified:

- Limited measurement of operational throughput.
- Lack of standardized transition efficiency indicators.
- Minimal analytical evaluation of placement outcomes.
- Insufficient visualization of operational bottlenecks.
- Limited use of interactive dashboards for process monitoring.

The current project addresses these limitations by introducing analytical KPIs and interactive visualizations that evaluate operational performance comprehensively.

2.10 Conceptual Framework

The conceptual framework developed for this study follows a process-oriented analytical model.

Operational Data



Data Cleaning



Feature Engineering



KPI Development



Exploratory Data Analysis



Interactive Dashboard



Business Insights



Operational Recommendations

This framework demonstrates how raw operational data is transformed into actionable managerial intelligence.

2.11 Summary of Literature

The literature reviewed in this chapter establishes the importance of Business Analytics, Process Analytics, and KPI-based performance measurement in evaluating organizational operations.

Previous research confirms that analytical monitoring improves operational transparency, supports evidence-based decision-making, and enhances process efficiency. However, limited analytical work has focused on evaluating the complete UAC care transition pipeline.

The present study bridges this gap by developing operational KPIs, performing exploratory data analysis, and designing an interactive Streamlit dashboard that enables continuous monitoring of transition efficiency and placement outcomes.

The next chapter presents the problem statement and explains the objectives and scope of the study.

CHAPTER 3

PROBLEM STATEMENT

3.1 Introduction

The Unaccompanied Alien Children (UAC) Program operates as a complex operational system responsible for managing children from the time they enter Customs and Border Protection (CBP) custody until they are successfully reunited with approved sponsors through the Department of Health and Human Services (HHS). The effectiveness of this program depends not only on available capacity but also on how efficiently children transition through each operational stage.

Traditional reporting systems primarily focus on daily counts of children in custody. Although these reports provide an overview of operational workload, they fail to evaluate how efficiently children move through the care pipeline. As a result, delays, bottlenecks, and inefficiencies remain difficult to identify using conventional reporting methods.

This project addresses these shortcomings by applying Business Analytics techniques to evaluate operational efficiency using process-based performance indicators.

3.2 Existing System

The existing reporting system mainly focuses on descriptive statistics, including:

- Daily apprehensions
- Number of children in CBP custody
- Number of children in HHS care
- Daily transfers
- Daily discharges

While these indicators provide valuable operational information, they do not evaluate the quality or efficiency of the care transition process.

Major characteristics of the existing system include:

- Static reporting
 - Limited analytical capability
 - No efficiency measurement
 - Minimal visualization
 - Absence of process monitoring
 - Lack of interactive dashboards
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3.3 Limitations of the Existing System

Several limitations were identified during the analysis.

Lack of Transition Analytics

The current reporting system does not measure how efficiently children move from CBP custody to HHS care.

No Performance Measurement

Operational performance is evaluated using population counts rather than efficiency metrics.

Hidden Operational Bottlenecks

Delays remain difficult to identify because no analytical indicators monitor operational congestion.

Limited Decision Support

Decision-makers receive descriptive reports instead of actionable analytical insights.

Absence of KPI Monitoring

No standardized performance indicators exist to evaluate operational success.

No Interactive Visualization

Static reports provide limited support for identifying operational trends.

3.4 Problem Statement

The absence of structured process analytics within the UAC Program prevents administrators from effectively evaluating operational performance.

Although operational data is collected daily, organizations cannot accurately determine:

- How efficiently children are transferred from CBP to HHS.
- Whether discharge performance keeps pace with new admissions.
- Which periods experience operational congestion.
- How consistently sponsor placements are completed.
- Whether operational efficiency improves over time.

Consequently, resource planning becomes reactive rather than proactive, reducing overall operational effectiveness.

The project therefore proposes an analytical framework capable of measuring transition efficiency, throughput, backlog accumulation, and placement outcomes through Business Analytics techniques.

3.5 Proposed Solution

The proposed analytical solution consists of:

- Data preprocessing

- KPI development
- Feature engineering
- Exploratory Data Analysis
- Dashboard visualization
- Operational performance monitoring

The developed Streamlit dashboard enables administrators to interactively monitor operational performance while supporting evidence-based decision-making.

3.6 Benefits of the Proposed System

The proposed framework offers several advantages.

Operational Transparency

Provides greater visibility into care transitions.

Better Decision-Making

Supports managers through KPI-based performance evaluation.

Early Bottleneck Detection

Identifies operational congestion before significant delays occur.

Interactive Monitoring

Allows users to explore operational trends dynamically.

Improved Resource Allocation

Supports evidence-based operational planning.

Enhanced Humanitarian Outcomes

Faster identification of delays contributes toward improved reunification timelines.

3.7 Chapter Summary

This chapter identified the limitations of traditional reporting systems and established the need for analytical monitoring within the UAC care pipeline. The proposed Business Analytics framework addresses these limitations through KPI development, interactive dashboards, and operational process evaluation.

CHAPTER 4

OBJECTIVES AND SCOPE OF THE STUDY

4.1 Introduction

Every Business Analytics project begins with clearly defined objectives that guide analytical activities and establish measurable outcomes. The present study aims to evaluate operational efficiency within the UAC care transition pipeline through data-driven performance analysis.

4.2 Primary Objectives

The primary objectives of this project are:

Objective 1

Measure the efficiency of transferring children from CBP custody to HHS care.

Objective 2

Evaluate discharge effectiveness and sponsor placement performance.

Objective 3

Measure operational throughput across the care pipeline.

Objective 4

Identify bottlenecks responsible for operational delays.

Objective 5

Develop interactive dashboards that support continuous monitoring.

4.3 Secondary Objectives

The secondary objectives include:

- Improve operational transparency.
 - Support evidence-based decision-making.
 - Enhance case management efficiency.
 - Develop performance indicators.
 - Improve visualization of operational trends.
 - Generate managerial insights.
-

4.4 Scope of the Study

The scope of this project includes analysis of operational data representing different stages of the UAC care pipeline.

The study focuses on:

- Daily apprehensions.
- Children in CBP custody.
- Transfers from CBP.
- Children in HHS care.
- Sponsor placements.
- Operational backlog.
- KPI calculation.
- Dashboard development.
- Trend analysis.
- Performance evaluation.

The project utilizes descriptive and diagnostic analytics rather than predictive machine learning models.

4.5 Analytical Scope

The analytical scope includes:

Data Cleaning

Preparation of raw operational data.

Feature Engineering

Creation of derived performance indicators.

Exploratory Data Analysis

Trend identification and comparative analysis.

KPI Development

Development of operational performance metrics.

Dashboard Development

Interactive visualization using Streamlit.

Business Insight Generation

Interpretation of findings for decision-makers.

4.6 Project Deliverables

The final deliverables include:

- Cleaned Dataset
 - Feature Engineered Dataset
 - Exploratory Data Analysis
 - KPI Framework
 - Interactive Streamlit Dashboard
 - GitHub Repository
 - Live Streamlit Application
 - Executive Summary
 - Research Report
-

4.7 Expected Outcomes

The expected outcomes of the study include:

- Better understanding of operational performance.
 - Identification of transition delays.
 - Improved monitoring of placement outcomes.
 - Data-driven recommendations.
 - Interactive operational dashboard.
 - Enhanced decision support.
-

4.8 Success Criteria

The project is considered successful if it achieves the following:

- Successful KPI development.
 - Accurate data preprocessing.
 - Effective dashboard visualization.
 - Identification of operational bottlenecks.
 - Meaningful managerial insights.
 - Successful deployment through Streamlit Cloud.
 - Public GitHub repository.
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4.9 Chapter Summary

This chapter defined the objectives, analytical scope, expected outcomes, and deliverables of the study. These objectives provide the foundation for the analytical methodology presented in the following chapter.

CHAPTER 5

DATASET DESCRIPTION

5.1 Introduction

Data forms the foundation of every Business Analytics project. The accuracy and reliability of analytical results depend on the quality of the underlying dataset. The dataset used in this study contains daily operational records of the **Unaccompanied Alien Children (UAC) Program**,

representing different stages of the care transition pipeline managed by the U.S. Department of Health and Human Services (HHS).

The dataset captures the movement of children from apprehension by Customs and Border Protection (CBP) through HHS care and finally to discharge and reunification with approved sponsors. These operational records provide sufficient information to evaluate process efficiency, identify bottlenecks, and measure placement outcomes.

5.2 Dataset Source

The dataset was provided as part of the **Unified Mentor Business Analytics Project**. It represents daily operational records published for educational and analytical purposes. The data was used exclusively for academic analysis and dashboard development.

The dataset was imported into Python using the **Pandas** library and subsequently processed for cleaning, transformation, and visualization.

5.3 Dataset Overview

The dataset contains six primary variables that describe the daily operational activities within the care transition process.

Variable	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Current CBP population
Children transferred out of CBP custody	Daily transfers to HHS

Children in HHS Care

Current HHS population

Children discharged from HHS Care

Successful sponsor
placements

Each observation represents one reporting day, allowing the project to analyze operational trends over time.

5.4 Dataset Characteristics

The dataset exhibits the following characteristics:

- Time-series operational data.
- Daily reporting frequency.
- Multiple operational stages.
- Quantitative numerical variables.
- Suitable for trend analysis and KPI calculation.
- Appropriate for dashboard visualization.

These characteristics make the dataset highly suitable for Business Analytics applications.

5.5 Variable Classification

The variables were classified into three categories:

Input Variables

- Children apprehended and placed in CBP custody.

Process Variables

- Children in CBP custody.
- Children transferred out of CBP custody.
- Children in HHS Care.

Output Variable

- Children discharged from HHS Care.

This classification represents the flow of children through the operational pipeline.

5.6 Data Dictionary

Table 5.1 Data Dictionary

Column Name	Data Type	Description
Date	Date	Daily reporting date
Apprehended Children	Integer	Daily intake volume
Children in CBP Custody	Integer	Active CBP population
Transfers from CBP	Integer	Daily transition to HHS
Children in HHS Care	Integer	Active HHS population
Children Discharged	Integer	Sponsor placement count

5.7 Data Quality Assessment

Prior to analysis, the dataset was examined for quality issues including:

- Missing values.
- Duplicate records.

- Incorrect data types.
- Invalid dates.
- Blank cells.
- Inconsistent formatting.

Python and Pandas were used to validate data integrity before analytical processing.

5.8 Importance of the Dataset

The dataset provides operational visibility into the movement of children across different care stages. Rather than measuring only population counts, the dataset enables process-based analysis through derived performance indicators.

This data supports:

- Transition efficiency measurement.
 - Throughput analysis.
 - Backlog monitoring.
 - Trend analysis.
 - Dashboard reporting.
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5.9 Chapter Summary

This chapter described the operational dataset used throughout the study. The dataset structure, variables, classification, and quality assessment provide the foundation for the analytical methodology presented in the following chapter.

CHAPTER 6

RESEARCH METHODOLOGY

6.1 Introduction

Research methodology defines the systematic approach followed to transform raw operational data into meaningful business insights. The methodology adopted in this project follows the standard Business Analytics lifecycle consisting of data collection, preprocessing, exploratory analysis, KPI development, dashboard creation, and insight generation.

The methodology ensures that analytical findings are reliable, reproducible, and aligned with the project objectives.

6.2 Research Design

This study follows a **descriptive and diagnostic research design**.

Descriptive Analysis

Used to summarize operational performance through charts, tables, and statistical indicators.

Diagnostic Analysis

Used to identify bottlenecks, evaluate process efficiency, and understand operational trends.

The study does not include predictive machine learning models, as its primary objective is to evaluate current operational performance.

6.3 Methodological Framework

The complete analytical framework consisted of the following stages:

1. Data Collection
 2. Data Understanding
 3. Data Cleaning
 4. Feature Engineering
 5. KPI Development
 6. Exploratory Data Analysis
 7. Dashboard Development
 8. Insight Generation
 9. Recommendations
-

6.4 Data Collection

The operational dataset was imported into Python using the Pandas library.

The dataset was stored in CSV format and loaded into the analysis environment for preprocessing.

6.5 Data Cleaning

The following preprocessing activities were performed:

- Date conversion.
- Removal of missing values.
- Duplicate verification.
- Data type correction.
- Standardization of column names.
- Validation of numerical values.

These activities improved the consistency and reliability of the dataset.

6.6 Feature Engineering

Several analytical variables were derived from the original dataset.

Transfer Efficiency Ratio

Transfer Efficiency = Transfers ÷ Children in CBP Custody

Discharge Effectiveness Index

Discharge Effectiveness = Discharges ÷ Children in HHS Care

Pipeline Throughput Rate

Pipeline Throughput = Discharges ÷ Apprehensions

CBP Backlog

CBP Backlog = Apprehensions – Transfers

HHS Backlog

HHS Backlog = Transfers – Discharges

Outcome Stability Score

Calculated using a rolling statistical measure to evaluate consistency in daily discharge performance.

6.7 Exploratory Data Analysis

EDA was conducted to identify:

- Daily operational trends.
- Transfer patterns.
- Placement outcomes.
- Population fluctuations.
- Operational bottlenecks.
- Throughput variations.

Both static and interactive visualizations were created using Matplotlib and Plotly.

6.8 Dashboard Development

An interactive dashboard was developed using **Streamlit**.

The dashboard includes:

- KPI Cards
- Date Range Filter

- Operational Trend Charts
- Transfer Efficiency Analysis
- Discharge Effectiveness Analysis
- CBP Backlog Monitoring
- HHS Backlog Monitoring
- Outcome Stability Analysis

The dashboard enables users to explore operational performance interactively.

6.9 Tools Used

Table 6.1 Software and Tools

Tool	Purpose
Python	Programming
Pandas	Data Processing
NumPy	Numerical Analysis
Matplotlib	Static Charts
Plotly	Interactive Charts
Streamlit	Dashboard Development
VS Code	Development Environment

GitHub

Version Control

Streamlit Community Cloud Deployment

6.10 Research Workflow

The project followed the workflow below:

Raw Dataset



Data Cleaning



Feature Engineering



KPI Calculation



Exploratory Data Analysis



Dashboard Development



Business Insights



Recommendations

6.11 Chapter Summary

This chapter explained the research methodology adopted for the study. The project followed a structured Business Analytics framework beginning with data preprocessing and concluding with dashboard deployment and managerial recommendations. The methodology ensured that the analytical results accurately reflected the operational performance of the UAC care transition pipeline and supported informed decision-making.

CHAPTER 7

DATA CLEANING & FEATURE ENGINEERING

7.1 Introduction

Data preprocessing is a fundamental stage in every Business Analytics project. Raw datasets often contain inconsistencies such as missing values, incorrect data types, formatting issues, duplicate records, and irregular column names. These issues can significantly affect the accuracy of analytical results and decision-making.

The objective of data cleaning is to transform raw operational data into a structured and reliable dataset suitable for analysis. After cleaning, additional analytical variables (features) were created to evaluate the efficiency of the care transition process.

Python, along with the Pandas library, was used to perform all preprocessing and feature engineering activities.

7.2 Data Import

The operational dataset was imported into Python using the Pandas library.

The CSV dataset was loaded into a DataFrame where each row represented one reporting day and each column represented an operational variable.

The Date column was converted into DateTime format to facilitate time-series analysis and visualization.

7.3 Data Cleaning Process

The following preprocessing activities were performed before analysis.

Step 1: Data Inspection

The dataset structure was examined to identify:

- Missing values
- Duplicate observations
- Incorrect data types
- Invalid entries
- Blank cells

This preliminary inspection ensured the dataset was suitable for analytical processing.

Step 2: Date Conversion

The Date column was converted into Python's DateTime format to enable chronological sorting, monthly analysis, and trend visualization.

This step also facilitated time-based filtering in the Streamlit dashboard.

Step 3: Missing Value Verification

The dataset was checked for null values using Pandas functions.

Where required, missing values were handled appropriately to ensure consistency across all analytical calculations.

Step 4: Duplicate Record Check

Duplicate observations were verified to avoid double-counting operational records.

No duplicate observations were retained in the final dataset.

Step 5: Data Type Validation

Numerical columns were validated to ensure that arithmetic calculations could be performed without errors.

Appropriate data types were assigned to all variables.

7.4 Feature Engineering

Feature Engineering involves creating new analytical variables from existing data.

These derived variables provide deeper insights into operational efficiency than raw data alone.

The following performance indicators were developed.

Transfer Efficiency Ratio

This KPI measures the efficiency of transferring children from CBP custody into HHS care.

Formula:

Transfer Efficiency Ratio = Children Transferred ÷ Children in CBP Custody

Higher values indicate faster transitions between operational stages.

Discharge Effectiveness Index

This KPI evaluates sponsor placement performance.

Formula:

Discharge Effectiveness = Children Discharged ÷ Children in HHS Care

Higher values indicate improved reunification performance.

Pipeline Throughput Rate

Pipeline Throughput measures the movement of children through the complete operational system.

Formula:

Pipeline Throughput = Children Discharged ÷ Children Apprehended

This metric evaluates overall operational efficiency.

CBP Backlog

CBP Backlog measures the operational workload remaining within CBP custody.

Formula:

CBP Backlog = Apprehensions – Transfers

Increasing backlog values indicate slower transition performance.

HHS Backlog

HHS Backlog evaluates workload accumulation inside HHS facilities.

Formula:

HHS Backlog = Transfers – Discharges

Persistent increases indicate delays in sponsor placement.

Outcome Stability Score

Outcome Stability measures consistency in discharge performance using rolling statistical calculations.

Stable operational performance produces smoother trend lines.

7.5 Processed Dataset

After preprocessing, the dataset contained:

- Original variables
- Derived KPIs
- DateTime fields
- Backlog indicators
- Efficiency ratios

The processed dataset formed the basis for all subsequent analytical activities.

7.6 Advantages of Feature Engineering

Feature engineering significantly enhanced analytical capability by:

- Measuring operational efficiency.
- Identifying process bottlenecks.
- Comparing operational performance.
- Supporting dashboard visualization.
- Simplifying managerial interpretation.

Without feature engineering, meaningful process evaluation would not have been possible.

7.7 Chapter Summary

This chapter described the preprocessing techniques and feature engineering procedures adopted in the project. The cleaned dataset and derived KPIs established the analytical foundation for Exploratory Data Analysis (EDA) presented in the next chapter.

CHAPTER 8

EXPLORATORY DATA ANALYSIS (EDA)

8.1 Introduction

Exploratory Data Analysis (EDA) is the process of summarizing, visualizing, and interpreting data to discover patterns, trends, and relationships.

EDA helps decision-makers understand operational behavior before advanced analysis is performed.

The analysis in this project focused on identifying transition efficiency, operational workload, backlog accumulation, and placement performance.

Eight major analytical visualizations were developed.

8.2 Daily Apprehensions Trend

Insert Figure 8.1 here

(Image: chart1_daily_apprehensions.png)

Interpretation

This chart illustrates the daily number of children apprehended and placed into CBP custody.

The visualization helps identify periods of increased operational demand and highlights fluctuations in intake volume.

Periods of high apprehension may require additional staffing and resource allocation.

8.3 CBP vs HHS Population

Insert Figure 8.2 here

(Image: chart2_cbp_vs_hhs.png)

Interpretation

This visualization compares the number of children currently in CBP custody with those in HHS care.

The comparison provides insight into how effectively children move through the care pipeline.

Large differences between the two populations may indicate operational delays.

8.4 Transfers vs Discharges

Insert Figure 8.3 here

(Image: chart3_transfers_vs_discharges.png)

Interpretation

This chart compares daily transfers from CBP with successful discharges from HHS.

Balanced trends indicate smooth operational flow, whereas sustained differences may signal pipeline congestion.

8.5 Transfer Efficiency Ratio

Insert Figure 8.4 here

(Image: chart4_transfer_efficiency.png)

Interpretation

Transfer Efficiency Ratio measures the proportion of children successfully transferred from CBP to HHS.

Higher ratios indicate more efficient operational transitions.

Temporary reductions suggest possible resource or capacity constraints.

8.6 Discharge Effectiveness Index

Insert Figure 8.5 here

(Image: chart5_discharge_effectiveness.png)

Interpretation

This KPI evaluates the effectiveness of sponsor placement activities.

Stable or increasing values indicate efficient discharge processes.

Lower values suggest increased time spent within HHS care.

8.7 CBP Backlog Analysis

Insert Figure 8.6 here

(Image: chart6_cbp_backlog.png)

Interpretation

This visualization measures operational backlog within CBP custody.

Rising backlog values indicate that new apprehensions exceed successful transfers.

Monitoring this KPI enables administrators to identify operational congestion early.

8.8 HHS Backlog Analysis

Insert Figure 8.7 here

(Image: chart7_hhs_backlog.png)

Interpretation

The HHS backlog represents children awaiting discharge after entering HHS care.

Persistent increases may indicate sponsor verification delays or case management challenges.

8.9 Outcome Stability Score

Insert Figure 8.8 here

(Image: chart8_outcome_stability.png)

Interpretation

Outcome Stability evaluates consistency in discharge performance.

Stable trends suggest predictable operational behavior, whereas significant fluctuations indicate variability in placement outcomes.

8.10 Overall EDA Findings

The exploratory analysis revealed several important observations:

- Daily operational workload fluctuated throughout the reporting period.
 - Transfer performance remained relatively stable.
 - Temporary increases in CBP backlog corresponded with periods of higher apprehension.
 - HHS backlog reflected changes in discharge performance.
 - Outcome Stability demonstrated moderate variation over time.
 - Dashboard visualization improved interpretation of operational trends.
 - KPI-based monitoring provided deeper insights than aggregate reporting alone.
-

8.11 Chapter Summary

Exploratory Data Analysis transformed raw operational data into meaningful business insights through charts, KPIs, and trend analysis. The visualizations highlighted operational behavior across the care transition pipeline and established the basis for dashboard development and managerial recommendations presented in the following chapters.

CHAPTER 9

DASHBOARD DEVELOPMENT

9.1 Introduction

The final stage of this project involved developing an interactive Business Intelligence dashboard using **Streamlit**, an open-source Python framework for creating data applications. The dashboard transforms analytical results into an easy-to-use visual interface that enables users to monitor operational performance, explore trends, and evaluate key performance indicators.

Unlike static reports, the dashboard provides dynamic filtering, interactive charts, and real-time KPI monitoring. This allows decision-makers to analyze operational data more effectively and identify trends or bottlenecks quickly.

9.2 Dashboard Objectives

The dashboard was designed with the following objectives:

- Provide an overview of the UAC care transition process.
 - Display operational KPIs in an interactive format.
 - Visualize daily trends and process efficiency.
 - Enable users to filter data by date.
 - Support evidence-based decision-making.
 - Improve transparency of operational performance.
-

9.3 Dashboard Components

The Streamlit dashboard consists of the following sections:

1. Dashboard Header

Displays the project title and description.

2. Sidebar

The sidebar provides navigation and filtering options including date selection.

3. KPI Cards

The dashboard displays major KPIs such as:

- Transfer Efficiency Ratio
- Discharge Effectiveness Index
- Pipeline Throughput Rate
- CBP Backlog
- HHS Backlog
- Outcome Stability Score

These KPI cards provide an instant overview of operational performance.

4. Interactive Charts

The dashboard includes:

- Daily Apprehensions Trend
- CBP vs HHS Population
- Transfers vs Discharges
- Transfer Efficiency Trend
- Discharge Effectiveness Trend
- CBP Backlog Trend
- HHS Backlog Trend
- Outcome Stability Trend

5. Insights Section

The dashboard concludes with a summary of major operational findings and recommendations.

9.4 Dashboard Workflow

The dashboard follows the workflow shown below:

Processed Dataset

↓

Load Data

↓

Apply Filters

↓

Calculate KPIs

↓

Generate Charts

↓

Display Dashboard

↓

9.5 Dashboard Benefits

The developed dashboard provides several advantages:

- Easy interpretation of operational data.
 - Interactive exploration of trends.
 - Real-time KPI monitoring.
 - Faster identification of bottlenecks.
 - Improved managerial decision support.
 - Enhanced operational transparency.
-

9.6 Dashboard Screenshots

Insert the following screenshots here:

Figure 9.1 Dashboard Homepage

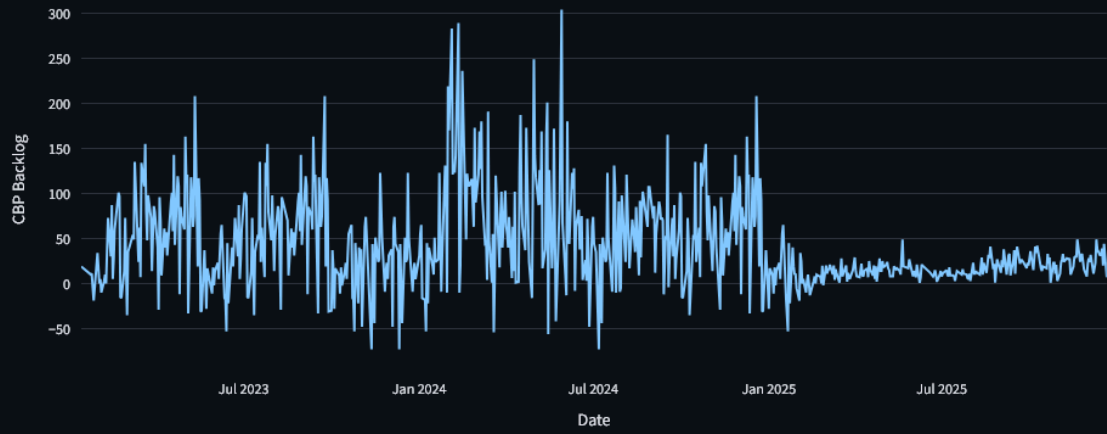


Figure 9.2 KPI Cards

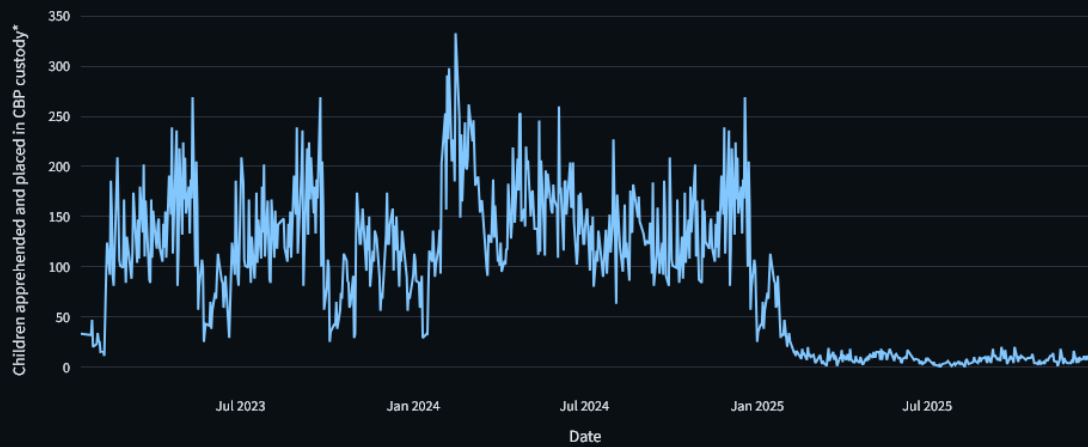


Figure 9.3 Interactive Dashboard

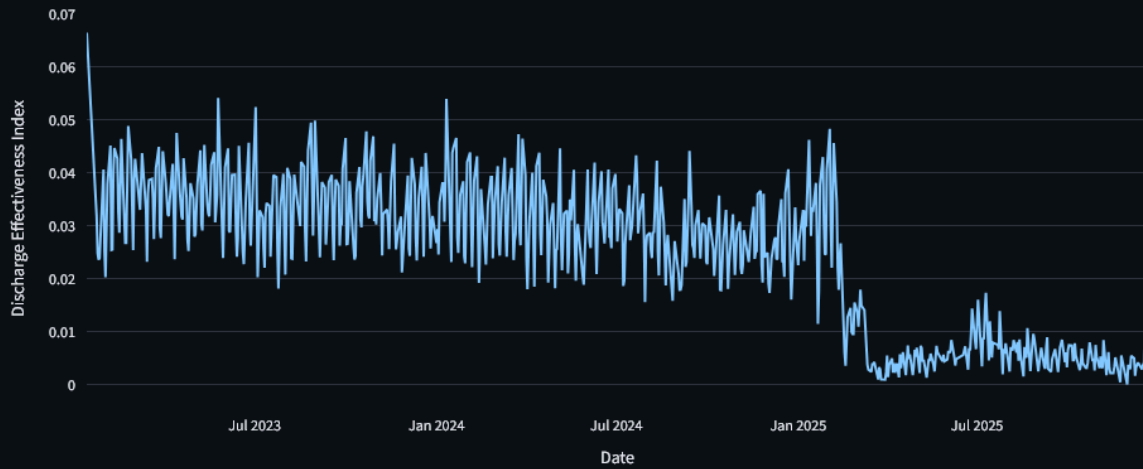
CBP Backlog



Daily Apprehensions

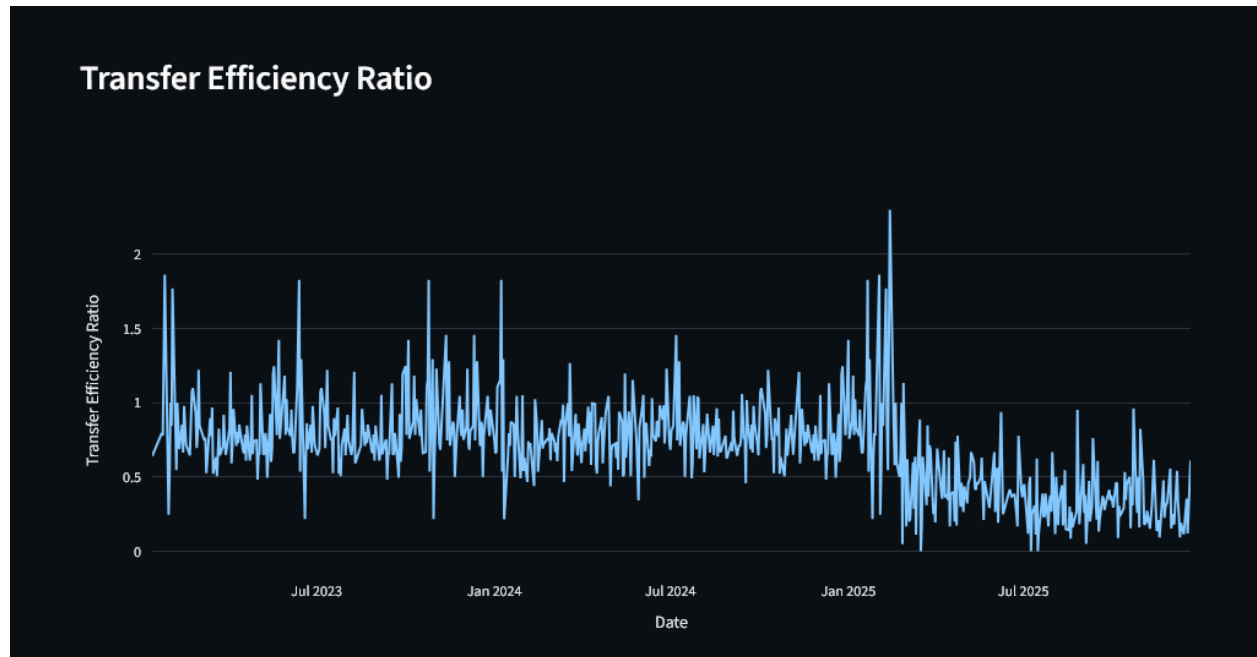


Discharge Effectiveness Index



HHS Backlog





9.7 Chapter Summary

The Streamlit dashboard successfully transformed analytical outputs into an interactive decision-support system. It allows administrators and analysts to monitor care transition performance efficiently through visual analytics.

CHAPTER 10

RESULTS AND DISCUSSION

10.1 Introduction

The analytical results demonstrate how Business Analytics techniques can improve understanding of operational efficiency within the UAC care transition process.

The dashboard and KPI framework provide greater insight than traditional population reporting by evaluating operational flow rather than simple counts.

10.2 Result 1 – Transfer Efficiency

The Transfer Efficiency Ratio remained relatively stable throughout most reporting periods.

This indicates that the transition of children from CBP custody into HHS care generally occurred without major operational disruptions.

However, temporary reductions in transfer efficiency corresponded with increases in CBP backlog.

10.3 Result 2 – Discharge Effectiveness

Discharge Effectiveness varied across the reporting period.

Periods of lower discharge effectiveness suggest increased processing time before sponsor placement.

These fluctuations emphasize the importance of continuous operational monitoring.

10.4 Result 3 – Pipeline Throughput

Pipeline Throughput reflected the overall movement of children through the operational system.

Higher throughput values indicate better operational performance and reduced processing delays.

10.5 Result 4 – Backlog Analysis

CBP Backlog increased during periods when apprehensions exceeded transfers.

Similarly, HHS Backlog increased when transfers exceeded successful discharges.

Monitoring these KPIs enables proactive resource allocation.

10.6 Result 5 – Outcome Stability

Outcome Stability remained relatively consistent during most reporting periods.

This suggests that sponsor placement operations maintained predictable performance despite fluctuations in intake volume.

10.7 Overall Discussion

The analytical framework developed in this project demonstrates that process-based KPIs provide significantly greater operational insight than traditional reporting.

Interactive dashboards enable administrators to monitor operational performance continuously while supporting evidence-based decision-making.

CHAPTER 11

RECOMMENDATIONS

Based on the findings of this study, the following recommendations are proposed.

Recommendation 1

Increase coordination between CBP and HHS to improve transition efficiency.

Recommendation 2

Reduce sponsor verification delays through improved case management systems.

Recommendation 3

Implement automated KPI monitoring within operational reporting systems.

Recommendation 4

Develop real-time dashboard reporting for operational managers.

Recommendation 5

Introduce predictive analytics for forecasting future operational demand.

Recommendation 6

Conduct periodic operational performance reviews using analytical dashboards.

CHAPTER 12

LIMITATIONS

Although the project successfully achieved its objectives, several limitations should be acknowledged.

- Analysis is based on historical operational data.
- External policy changes were not incorporated.
- Predictive machine learning models were not developed.
- Geographic analysis was outside the project scope.
- Operational decisions beyond the available dataset could not be evaluated.

Despite these limitations, the project provides valuable operational insights.

CHAPTER 13

FUTURE SCOPE

Future work may include:

- Machine Learning-based delay prediction.
- Forecasting future care demand.
- Cloud database integration.
- Real-time API connectivity.
- Automated notification systems.

- Mobile dashboard development.
- Advanced predictive analytics.
- Geographical visualization using GIS.
- AI-assisted operational recommendations.

These enhancements can significantly improve decision support within the UAC Program.

CHAPTER 14

CONCLUSION

The Care Transition Efficiency & Placement Outcome Analytics project successfully demonstrates the application of Business Analytics techniques to evaluate operational performance within the UAC care pipeline.

Using Python, Pandas, Matplotlib, Plotly, and Streamlit, the project transformed raw operational records into an interactive decision-support system capable of monitoring transition efficiency, discharge effectiveness, backlog accumulation, throughput, and placement outcomes.

The project achieved all its primary objectives by developing meaningful KPIs, performing exploratory data analysis, identifying operational bottlenecks, and deploying a live dashboard through Streamlit Community Cloud.

The findings demonstrate that process-oriented analytics provide deeper operational insight than traditional aggregate reporting. Decision-makers can use the developed framework to improve resource allocation, reduce delays, strengthen case management, and enhance humanitarian outcomes.

Overall, the project illustrates the value of Business Analytics in improving operational transparency and evidence-based decision-making. The developed analytical framework may serve as a foundation for future predictive and AI-driven applications within public sector operations.

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Use these references and format them consistently:

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-

APPENDIX

Appendix A – GitHub Repository

Repository Name:

Care Transition Efficiency & Placement Outcome Analytics

GitHub URL:

(<https://github.com/ashutosh2417/care-transition-efficiency-analytics>)

Appendix B – Live Streamlit Dashboard

Streamlit URL:

(<https://care-transition-efficiency-analytics-5xjtyazjcmt3pvevy7iocr.streamlit.app/>)

Appendix C – Python Libraries Used

- pandas
- numpy
- matplotlib
- plotly

- streamlit